

2AMC15 – Reinforcement Learning in Practice

2025/2026, Quartile 4

A. Report (30 points)

- Introduction (5 points)
 - + clearly states and motivates the three chosen algorithms from each category: (i) dynamic programming, (ii) Monte-Carlo method, and (iii) temporal difference learning.
 - +++ correctly explains how each algorithm works in a concise way.
 - + provides an overall technical comparison between all three.
- Experimental setups (10 points)
 - ++ clearly describes the basic (or shared) settings for the performed experiments, including the simulation environment, reward function, default (hyper-)parameters, grid generation, etc.
 - +++ includes and explains at least six experimental setups using various (hyper-)parameter configurations of the agent and/or environment for performance evaluation (must include *A1_grid.npy*).
 - +++ provides adequate justification for all six cases (or more) on why these experiments are important and relevant.
 - ++ describes and justifies at least two performance metrics used to evaluate the algorithms.
- Experimental results (10 points)
 - ++++++ presents the obtained results for all six cases using clear visualization (plots/tables/etc.), and provides a thorough analysis, comparing the implemented algorithms, for various setups and performance metrics.
 - +++ provides a detailed discussion on the observed performance of different methods in relation with different parameters/settings.
 - + presents the best performing approach on *A1_grid.npy*, its settings, and draws the optimal obtained policy.
- Conclusions (5 points)
 - +++ provides a conclusion regarding each implemented method and the configuration that they work best.
 - ++ provides a take-home message from the overall assignment, including a discussion on the validity of the solution and how it can be improved in the future.

B. Code (10 points)

- 0 point: there is no submitted zip/folder of the code, or it is just the same as the template code from the course, or the Python files are in a wrong format (other than .py).
- 2 points: there is no clear readme/instruction on how to work with the code or the code does not run and gives multiple errors (including path error, package not found, etc.).
- 4 points: the code runs without errors given the instruction in the readme file, but the implementation of the methods is incorrect or incomplete (not all three methods are coded or they are not correct).
- 6 points: all methods are correctly implemented, and the code runs without errors/problems, but the results illustrated in the report are not fully reproducible.
- 8 points: the code is clear, runs correctly, and reproduces the main results in the report, but it might be slow, difficult to configure/interact/understand, etc. and still could be improved.
- 10 points: the code is fully reproducible, well-documented, efficient, clean, and can be used easily as a standalone tool by potential stakeholders.

C. Deductions (-5 points)

- -2 points: the submission is not entirely anonymous.
- -2 points: the report does not adhere to the provided latex format.
- -1 point: the PDF file of the report is not submitted separately in the “Group Assignment 1” (not relevant for the reviewing assignment).